

HPE and HAR First-Round Result Summary

Blue indicates improvement over official X-Fi; red indicates degradation. HPE uses MPJPE/PA-MPJPE where lower is better. HAR uses accuracy where higher is better.

Completion Audit

Project	Done	Total
HPE	8	46
HAR	24	25

HPE Official Comparison

Method	Split	Source	Modality	Ours MPJPE	X-Fi MPJPE	Δ	PA
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk	487.99	93.90	+394.09	+79.32
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth	234.48	101.80	+132.68	+36.28
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	lidar	1615.42	167.10	+1448.32	+284.32
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	mmwave	4196.64	127.40	+4069.24	+228.24
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	wifi-csi	4173.33	225.60	+3947.73	+272.73
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth	369.12	86.10	+283.02	+49.02
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+lidar	1072.74	93.00	+979.74	+153.74
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+mmwave	814.71	88.80	+725.91	+116.91
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+wifi-csi	2491.83	93.00	+2398.83	+275.83
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+lidar	285.80	102.50	+183.30	+39.30
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+mmwave	178.87	98.00	+80.87	+18.87
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+wifi-csi	197.61	101.80	+95.81	+25.81
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	lidar+mmwave	2157.14	109.80	+2047.34	+324.34
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	lidar+wifi-csi	4007.08	159.50	+3847.58	+281.58
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	mmwave+wifi-csi	4675.91	117.20	+4558.71	+210.71
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+lidar	634.60	84.80	+549.80	+64.80
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+mmwave	465.16	83.40	+381.76	+39.76
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+wifi-csi	100.71	85.30	+15.41	+3.41
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+lidar+mmwave	1035.05	88.40	+946.65	+180.65
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+lidar+wifi-csi	4736.11	93.00	+4643.11	+211.11
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+mmwave+wifi-csi	4769.71	88.50	+4681.21	+198.21
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+lidar+mmwave	310.49	96.00	+214.49	+28.49
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+lidar+wifi-csi	177.40	102.00	+75.40	+18.40
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+mmwave+wifi-csi	121.88	97.00	+24.88	+7.88
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	lidar+mmwave+wifi-csi	4360.45	107.40	+4253.05	+259.05
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+lidar+mmwave	593.34	83.50	+509.84	+45.84
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+lidar+wifi-csi	67.89	86.00	-18.11	-7.11
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+mmwave+wifi-csi	71.27	84.00	-12.73	-8.73
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+lidar+mmwave+wifi-csi	4876.08	88.60	+4787.48	+182.48
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	depth+lidar+mmwave+wifi-csi	112.99	97.60	+15.39	+3.39
VK-RCD-Teacher	random_split	teacher_all.combinations.csv	vk+depth+lidar+mmwave+wifi-csi	48.45	83.70	-35.25	-14.25
VK-RCD-Student-VK	random_split	all_combinations.csv	vk	92.66	93.90	-1.24	-17.24
VK-RCD-Student-VK	random_split	all_combinations.csv	depth	53.88	101.80	-47.92	-12.92
VK-RCD-Student-VK	random_split	all_combinations.csv	lidar	139.90	167.10	-27.20	-9.20
VK-RCD-Student-VK	random_split	all_combinations.csv	mmwave	114.98	127.40	-12.42	-11.42
VK-RCD-Student-VK	random_split	all_combinations.csv	wifi-csi	222.14	225.60	-3.46	+4.46
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth	51.65	86.10	-34.45	-13.45
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+lidar	73.18	93.00	-19.82	-17.82
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+mmwave	68.93	88.80	-19.87	-18.87
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+wifi-csi	88.11	93.00	-4.89	-17.89
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+lidar	53.00	102.50	-49.50	-13.50
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+mmwave	52.32	98.00	-45.68	-12.68
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+wifi-csi	53.54	101.80	-48.26	-12.26
VK-RCD-Student-VK	random_split	all_combinations.csv	lidar+mmwave	97.54	109.80	-12.26	-8.26
VK-RCD-Student-VK	random_split	all_combinations.csv	lidar+wifi-csi	134.11	159.50	-25.39	-9.39
VK-RCD-Student-VK	random_split	all_combinations.csv	mmwave+wifi-csi	108.54	117.20	-8.66	-5.66
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+lidar	51.10	84.80	-33.70	-14.70
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+mmwave	50.50	83.40	-32.90	-13.90
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+wifi-csi	51.63	85.30	-33.67	-13.67
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+lidar+mmwave	66.08	88.40	-22.32	-18.32
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+lidar+wifi-csi	70.68	93.00	-22.32	-17.32
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+mmwave+wifi-csi	66.69	88.50	-21.81	-18.81
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+lidar+mmwave	51.92	96.00	-44.08	-12.08
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+lidar+wifi-csi	52.86	102.00	-49.14	-12.14
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+mmwave+wifi-csi	52.13	97.00	-44.87	-12.87
VK-RCD-Student-VK	random_split	all_combinations.csv	lidar+mmwave+wifi-csi	94.20	107.40	-13.20	-8.20
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+lidar+mmwave	50.25	83.50	-33.25	-14.25
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+lidar+wifi-csi	51.05	86.00	-34.95	-14.95
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+mmwave+wifi-csi	50.41	84.00	-33.59	-14.59
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+lidar+mmwave+wifi-csi	64.22	88.60	-24.38	-18.38
VK-RCD-Student-VK	random_split	all_combinations.csv	depth+lidar+mmwave+wifi-csi	51.78	97.60	-45.82	-12.82
VK-RCD-Student-VK	random_split	all_combinations.csv	vk+depth+lidar+mmwave+wifi-csi	50.17	83.70	-33.53	-14.53

HAR Official Comparison

Method	Split	Source	Modality	Ours Acc.	X-Fi Acc.
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk	3.40	26.50
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth	83.71	48.10
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	lidar	3.36	52.70
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	mmwave	66.72	85.70
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth	83.50	45.30
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+lidar	3.91	35.20
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+mmwave	66.46	73.40
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+lidar	83.37	51.60
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+mmwave	95.76	79.80
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	lidar+mmwave	66.69	88.70
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+lidar	83.66	48.70
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+mmwave	95.66	70.70
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+lidar+mmwave	66.39	77.80
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+lidar+mmwave	95.64	80.50
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+lidar+mmwave	95.57	72.20
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk	65.31	26.50
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth	88.02	48.10
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	lidar	40.80	52.70
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	mmwave	85.88	85.70
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth	89.51	45.30
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+lidar	68.54	35.20
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+mmwave	92.75	73.40
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+lidar	88.52	51.60
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+mmwave	96.45	79.80
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	lidar+mmwave	89.51	88.70
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+lidar	89.62	48.70
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+mmwave	96.56	70.70
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+lidar+mmwave	92.73	77.80
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+lidar+mmwave	96.44	80.50
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+lidar+mmwave	96.56	72.20
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk	3.40	26.50
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth	83.71	48.10
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	lidar	3.45	52.70
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	mmwave	66.72	85.70
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth	83.50	45.30
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+lidar	3.85	35.20
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+mmwave	66.46	73.40
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+lidar	83.39	51.60
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+mmwave	95.76	79.80
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	lidar+mmwave	66.64	88.70
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+lidar	83.70	48.70
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+mmwave	95.66	70.70
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+lidar+mmwave	66.33	77.80
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+lidar+mmwave	95.62	80.50
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+lidar+mmwave	95.59	72.20
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk	60.94	26.50
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth	86.85	48.10
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	lidar	33.14	52.70
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	mmwave	83.13	85.70
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth	88.09	45.30
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+lidar	63.63	35.20
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+mmwave	91.05	73.40
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+lidar	87.24	51.60
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+mmwave	95.57	79.80
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	lidar+mmwave	85.92	88.70
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+lidar	88.62	48.70
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+mmwave	95.96	70.70
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+lidar+mmwave	91.23	77.80
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+lidar+mmwave	95.50	80.50
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+lidar+mmwave	95.99	72.20
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk	5.52	26.50
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth	85.59	48.10
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	lidar	2.65	52.70
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	mmwave	69.91	85.70
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth	86.48	45.30
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+lidar	4.58	35.20
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+mmwave	67.75	73.40
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+lidar	86.16	51.60
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+mmwave	95.54	79.80
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	lidar+mmwave	67.80	88.70
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+lidar	86.43	48.70
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+mmwave	95.50	70.70
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+lidar+mmwave	67.09	77.80
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+lidar+mmwave	95.42	80.50
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+lidar+mmwave	95.39	72.20